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  CONVEX-ROOF C_GME  -- METHOD-2 (Exponential Map)
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  Generated          : 2026-04-10  21:07
  System             : 1D Hubbard ring  (mode: uniform)

  SYSTEM PARAMETERS
  ------------------------------------------------
  Sites          L   = 3
  Electrons     Ne   = 3
  Hopping        t   = 1.0
  Interaction    U   = 4.0

  GROUND STATE
  ------------------------------------------------
  E0                 = -1.2749172176
  E1                 = -1.2749172176
  Gap  E1-E0         = 0.0000000000
  <S^2>              = 0.750000
  S                  = 0.5000
  <Sz>               = 0.250000
  Multiplicity       = 2.0
  Spin state         = Doublet (S=1/2)
  <D> total          = 0.230835
  <D> / site         = 0.076945

  ANALYSIS PARAMETERS
  ------------------------------------------------
  Clusters analysed  = 3
  Optimization Methods used       = 6
  Restarts / method  = 20

  OPTIMIZATION METHODS
  ------------------------------------------------
  1. L-BFGS-B
  2. Nelder-Mead
  3. Powell
  4. COBYLA
  5. Differential-Evo
  6. Simulated-Ann

  NOTATION
  ------------------------------------------------
  C_LB     = Lower bound  (paper coherence-witness, analytical)
  C_GME    = GME for pure rdm
  C_UB     = Upper bound  (eigendecomposition, analytical)
  C_GME_X  = Convex-roof estimate from optimiser X

  True C_GME satisfies:  C_LB <= C_GME <= C_UB

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  CLUSTER OVERVIEW  --  All Optimizers
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  | Cluster          |       Purity |         C_LB |        C_GME |         C_UB | C_GME_L-BFGS-B | C_GME_Nelder-M |   C_GME_Powell |   C_GME_COBYLA | C_GME_Diff-Evo |  C_GME_Sim-Ann | BestMethod   | Status |
  +------------------+--------------+--------------+--------------+--------------+----------------+----------------+----------------+----------------+----------------+----------------+--------------+--------+
  | (0, 1, 2)        |   1.00000000 |   0.00000000 |   0.70079342 |   0.70079342 | *   0.70079342 |     0.70079342 |     0.70079342 |     0.70079342 |     0.70079342 |     0.70079342 | L-BFGS-B     |    MIX |
  | (1, 2, 0)        |   1.00000000 |   0.00000000 |   0.70079342 |   0.70079342 | *   0.70079342 |     0.70079342 |     0.70079342 |     0.70079342 |     0.70079342 |     0.70079342 | L-BFGS-B     |    MIX |
  | (2, 0, 1)        |   1.00000000 |   0.00000000 |   0.70079342 |   0.70079342 | *   0.70079342 |     0.70079342 |     0.70079342 |     0.70079342 |     0.70079342 |     0.70079342 | L-BFGS-B     |    MIX |
  +------------------+--------------+--------------+--------------+--------------+----------------+----------------+----------------+----------------+----------------+----------------+--------------+--------+
  * = best (lowest) C_GME for that cluster

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  Optimizers SUMMARY  (aggregated across all clusters)
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  Optimizer            Mean C_GME    Std Dev         Best        Worst   Total(s)  Rank
  ------------------------------------------------------------------------------------
  L-BFGS-B             0.70079342   0.000000   0.70079342   0.70079342       0.00     1
  Nelder-Mead          0.70079342   0.000000   0.70079342   0.70079342       0.00     2
  Powell               0.70079342   0.000000   0.70079342   0.70079342       0.00     3
  COBYLA               0.70079342   0.000000   0.70079342   0.70079342       0.00     4
  Differential-Evo     0.70079342   0.000000   0.70079342   0.70079342       0.00     5
  Simulated-Ann        0.70079342   0.000000   0.70079342   0.70079342       0.00     6

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  TIMING ANALYSIS
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  Optimizer            Total(s)   Avg/cluster(s)  Fraction(%)  Rank(quality)
  ------------------------------------------------------------------------------
  L-BFGS-B                 0.00             0.00          0.0              1
  Nelder-Mead              0.00             0.00          0.0              2
  Powell                   0.00             0.00          0.0              3
  COBYLA                   0.00             0.00          0.0              4
  Differential-Evo         0.00             0.00          0.0              5
  Simulated-Ann            0.00             0.00          0.0              6
  ------------------------------------------------------------------------------
  TOTAL                    0.00


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================================================================================================================================================
  CLUSTER DETAIL  --  Sites (0, 1, 2)
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  n (body order) = 3
  Purity Tr(r^2) = 1.0000000000
  Rank           = 1
  C_LB (lower)   = 0.0000000000
  C_GME   = 0.7007934184
  C_UB (upper)   = 0.7007934184
  Best C_GME (optimizer)     = 0.7007934184  [L-BFGS-B]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2)                 0.5547938505
  (1,)                 (0, 2)                 0.7544442924
  (2,)                 (0, 1)                 0.7544442924

  Method results
  ------------------------------------------------------------------------------
  Method                       C_GME   Time(s)  Restarts   Evals        Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.7007934184      0.00         0       0     0.00e+00
  Nelder-Mead          0.7007934184      0.00         0       0     0.00e+00
  Powell               0.7007934184      0.00         0       0     0.00e+00
  COBYLA               0.7007934184      0.00         0       0     0.00e+00
  Differential-Evo     0.7007934184      0.00         0       0     0.00e+00
  Simulated-Ann        0.7007934184      0.00         0       0     0.00e+00

  Pure-state decomposition  [L-BFGS-B]
  ------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------
     1   1.0000000000     0.7007934184   0.7007934184
  ------------------------------------------------------
   Sum                                  0.7007934184
  ( best C_GME = 0.7007934184)

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================================================================================================================================================
  CLUSTER DETAIL  --  Sites (1, 2, 0)
================================================================================================================================================

  n (body order) = 3
  Purity Tr(r^2) = 1.0000000000
  Rank           = 1
  C_LB (lower)   = 0.0000000000
  C_GME   = 0.7007934184
  C_UB (upper)   = 0.7007934184
  Best C_GME (optimizer)     = 0.7007934184  [L-BFGS-B]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2)                 0.6907618571
  (1,)                 (0, 2)                 0.5547938505
  (2,)                 (0, 1)                 0.5547938505

  Method results
  ------------------------------------------------------------------------------
  Method                       C_GME   Time(s)  Restarts   Evals        Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.7007934184      0.00         0       0     0.00e+00
  Nelder-Mead          0.7007934184      0.00         0       0     0.00e+00
  Powell               0.7007934184      0.00         0       0     0.00e+00
  COBYLA               0.7007934184      0.00         0       0     0.00e+00
  Differential-Evo     0.7007934184      0.00         0       0     0.00e+00
  Simulated-Ann        0.7007934184      0.00         0       0     0.00e+00

  Pure-state decomposition  [L-BFGS-B]
  ------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------
     1   1.0000000000     0.7007934184   0.7007934184
  ------------------------------------------------------
   Sum                                   0.7007934184
  ( best C_GME = 0.7007934184)

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================================================================================================================================================
  CLUSTER DETAIL  --  Sites (2, 0, 1)
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  n (body order) = 3
  Purity Tr(r^2) = 1.0000000000
  Rank           = 1
  C_LB (lower)   = 0.0000000000
  C_GME   = 0.7007934184
  C_UB (upper)   = 0.7007934184
  Best C_GME (optimizer)     = 0.7007934184  [L-BFGS-B]
  Bounds : C_LB <= C_GME <= C_UB?     = YES

  Bipartition purities
  ------------------------------------------------------------
  A                    B                      Tr(rho_A^2)
  ------------------------------------------------------------
  (0,)                 (1, 2)                 0.7544442924
  (1,)                 (0, 2)                 0.6907618571
  (2,)                 (0, 1)                 0.6907618571

  Method results
  ------------------------------------------------------------------------------
  Method                       C_GME   Time(s)  Restarts   Evals        Spread
  ------------------------------------------------------------------------------
  L-BFGS-B             0.7007934184      0.00         0       0     0.00e+00
  Nelder-Mead          0.7007934184      0.00         0       0     0.00e+00
  Powell               0.7007934184      0.00         0       0     0.00e+00
  COBYLA               0.7007934184      0.00         0       0     0.00e+00
  Differential-Evo     0.7007934184      0.00         0       0     0.00e+00
  Simulated-Ann        0.7007934184      0.00         0       0     0.00e+00

  Pure-state decomposition  [L-BFGS-B]
  ------------------------------------------------------
     #     weight p_i   C_GME(|psi_i>)      p_i*C_GME
  ------------------------------------------------------
     1   1.0000000000     0.7007934184   0.7007934184
  ------------------------------------------------------
   Sum                                   0.7007934184
  ( best C_GME = 0.7007934184)

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